



How public information affects asymmetrically informed lenders: Evidence from a credit registry reform

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ABSTRACT

We exploit exogenous variation in firm's public information available to banks to empirically evaluate the importance of adverse selection in the credit market using a Pakistani banking reform that reduced public information. Originally, the central bank published credit information about the firm *and* (aggregate) credit information about the firm's group. After the reform, the central bank stopped providing the aggregate group-level information. We construct a measure for the amount of information each lender has about a firm's group using the set of firm-bank lending pairs prior to the reform. We show those banks with private information about a firm lent relatively more to that firm than other, less-informed banks following the reform. Remarkably, this reduction in lending by less informed banks is true even for banks that had a preexisting relationship with the firm, suggesting that the strength of prior relationships does not eliminate the problem of imperfect information.

1. Introduction

Credit markets are crucial to economic growth, but asymmetric information may hinder their effectiveness. Jaffee and Russell (1976) and Stiglitz and Weiss (1981) theoretically show that adverse selection is a factor that can substantially constrain the effectiveness of credit markets, yet the extent to which this factor is a problem in credit markets remains largely unknown. It is possible, for example, that lenders face other binding constraints (such as those due to moral hazard) that make adverse selection less relevant.

The key empirical challenge to isolating the importance of adverse selection in the real world is that the variation in the individual observations is unlikely to be exogenous. For example, lending could drive the asymmetry of information rather than the reverse: one bank may know more about a firm than another bank simply because it has a lent to that firm in the past.

An innovative paper by Karlan and Zinman (2009) on adverse selection (and moral hazard) in credit markets tackles one aspect of this issue. Karlan et al. examine if exogenously lowering the interest rate changes selection of the borrower pool in ways that the lender cannot observe. While they are able to highlight borrower behavior, they

are still unable to observe changes in lender behavior in the presence of information asymmetries. In particular, they are not able to demonstrate that lenders react to these changing unobservables by altering the terms they offer—since the set of contracts offered by the lenders is fixed—which is the key mechanism behind credit market failure in the theoretical literature.

The purpose of this paper is to show that banks do change their lending behavior in reaction to a change in the distribution of information about a particular firm. In particular, among a group of differentially informed lenders, we ask: What is the effect of reducing public information about a borrower's creditworthiness?

A regulatory reform implemented by the State Bank of Pakistan (SBP) in April 2006 offers a unique opportunity to consider this question.¹ This reform exogenously reduced the amount of public information available to lenders about a firm's creditworthiness and did so in a way that varied across firm-lender pairs. Specifically, the change limited a lender's capacity to procure information about a firm's relationships to other firms.

Until April 2006, the SBP had supplemented credit information about prospective borrowers with information about that firm's "group." This group was defined as all other firms which shared at

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¹ The SBP announced its intention to implement this reform in May 2004, but details on how the reform would be implemented nor a detailed timeline was provided.

least one director with the borrowing firm.² But in April 2006, the SBP stopped defining a firm's group and in doing so stopped providing group-specific information (see section 3.1 for more detail).

Lenders value such information about the credit of other firms in a borrower's network because each firm's financial conditions will be linked. For example, business group owners may tunnel profits out of low-ownership stake firms, business groups may insure against idiosyncratic risks, or politically connected firms may be able to garner political favors for the entire set of related firms. In our study, we find strong evidence for linked financial performance of group firms. In our data set the unconditional probability of a corporate loan being overdue in December 2004 was only 2.6 percent. However, conditional on any firm in the firm's group being overdue, that firm was 6–7 percentage points more likely to also be overdue.

As anecdotal evidence for how the financial and economic relationships between firms in the same group can be tied, consider a Pakistani business group, the Dewan Group, that has at least 29 limited liability companies associated with the group. These firms span a very diverse set of industries, including automobile parts production, aviation, catering, cement, petroleum, textiles, and sugar mills. Financial and directorial information is available for 13 interconnected firms. In 2005, across these 13 firms, there are a total of 68 directorships, but due to the firms having common directors, there are actually only 28 unique directors. These companies have both strong interfirm relationships as well as similar financial performance. For instance, in 2005, none of the 13 companies had an overdue loan, yet, by 2010, 10 of these companies had loans that were overdue more than 365 days.

We take advantage of the natural experiment the SBP reform generated: We use a difference-in-difference methodology to estimate the causal effect of the reduction in public information on a bank's willingness to lend.³ In particular, we exploit variation in the impact of the policy across firm-bank pairs generated by each bank's other lending relationships. Suppose a firm borrowed from two different banks in December 2004. Following the reform, we predict that one bank—the informed bank—could have more information about the firm's actual liabilities, if other members of the firm's group also borrow from that bank. We then compare how the loans that the firm receives from the informed and the uninformed banks change after the reform. In addition, we examine whether the reform affected a firm's ability to access total credit by comparing loan amounts received by firms who had informed lenders and those who did not, before and after the policy reform.

Our main result shows that banks with private information about other firms in a firm's group lent more to that firm than other, less-informed lenders did, after the reform, both on an extensive margin (5.4 percentage points more likely to renew the firm's loan) and on an intensive margin (larger loans). This is the primary evidence that this reduction in public information amplified the problem of adverse selection in the credit market.

Second, those firms that borrowed from informed lenders were likely to borrow 11 to 14 percent more than those firms who did not have access to informed lenders. In other words, both the level of credit and its source were affected.

Third, using the distribution of firm-bank pairs and the strength of interfirm relations, we construct a measure for the quality of private information each bank possessed about a firm's group prior to the

reform. Following the reform, those lenders with greater private information about a firm's group were more likely to renew a firm's loan.⁴ This further supports the claim that a greater information differential between lenders leads to a greater disparity in the likelihood of a loan being renewed.

Fourth, there is substantial heterogeneity in the measured effect across firms: The smaller firms in a group were the most disadvantaged from the information change. There was no effect on the largest firms in a group, which suggests either that information about these borrowers was already pervasive or that the relative cost of procuring the information was lower.

Fifth, those firms which had negative information in their credit reports in December 2004 were the most affected by the change in information reporting. They were 18 percentage points more likely to renew their loan at an informed lender. Those firms with a poor credit history are likely to be the most risky for a bank, making public information all the more important for these firms. A firm may have overdue loans, but if the rest of the group is prompt in their repayment, it may signal a sufficiently creditworthy borrower. However, if both the firm and the group are overdue, this could signal wider systemic issues in the firm's creditworthiness.

The rest of the paper proceeds as follows: Section 2 reviews the relevant literature on credit markets, and section 3 describes the institutional background and explains the SBP's reasoning in instituting the change in lending policy. Section 4 outlines our econometric framework for analyzing the importance of information asymmetries across lenders and details our results. Section 5 outlines the effects of the reform on a firm's ability to procure credit, and section 6 presents a summary and concluding remarks for future research.

2. Literature review

This paper's main contribution is to empirically assess the impact of adverse selection in credit markets with differentially informed lenders. There is a long theoretical literature describing how asymmetric information in credit markets causes lenders to alter what contracts they offer. Compared to an environment with full information, this can lead to the misallocation of capital (Jaffee and Russell (1976); Stiglitz and Weiss (1981, 1983); De Meza and Webb (1987)), even to the complete unraveling of the credit market (Akerlof (1970)).

Karlan and Zinman's novel experiment separately identified adverse selection and moral hazard in microcredit but only found weak evidence for adverse selection effects. It is unclear if the small effects they find are due to the small loans in the microcredit sector, or because the experiment only used individuals who borrowed from the lender previously, or if adverse selection is not a key problem in lending. Ausubel (1999) uses a randomized trial which varied the contractual terms for a pre-approved credit card for 600,000 individuals. Contrary to Karlan and Zinman, Ausubel finds large effects of adverse selection in the credit card industry. We examine the effects of adverse selection in a market with much larger loans and in a setting where we expect the effect of adverse selection to be very different. Additionally, we examine how lenders react to the change in a set of observable characteristics—which is the key mechanism leading to credit market failure in the theoretical literature.

The classic literature on information problems in credit markets highlights the effect of asymmetric information on symmetrically informed lenders. Recent papers examine the theoretical and empirical effect of information asymmetries across lenders. Stroebel (2013)

² Firms within a group have complex interfirm relationships which subsequently have important economic implications. There is both theoretical and empirical evidence that interfirm relationships may be a mechanism for tax reduction (Desai and Dharmapala (2009)), tunneling (Bertrand et al., (2002)), risk sharing (Khwaja and Mian (2005a)), the efficient working of organizations (Williamson (1975; 1981)) and internal capital markets (Stein (1997); Almeida and Wolfenzon (2006); Gopalan et al., (2007)).

³ For brevity, in this paper we use "bank" to denote any financial institution.

⁴ Since the strength of interfirm ties vary, we can construct various measures for the quality of private information a bank may possess about a firm dependent on who the bank lends to within that firm's group, and the relative strength of those interfirm relationships.

theoretically models the interest rates a borrower receives when mortgage lenders are differentially informed about the borrower's collateral. He shows empirically that the return is higher for more informed lenders. Moreover, due to the winner's curse, less informed lenders charge higher interest rates when competing against more informed lenders.

Consistent with Stroebe's results, we find the more informed lenders to be better off. Yet our environment has richer intertemporal variation that allows us to address different issues. First, while Stroebe relies on pre-existing differences in whether the lender also built the property, we identify the effect of differential information exploiting an exogenous change in public information reporting. Second, because firms in our setting have multiple loans, we can estimate the effect of differential information across lenders for the *same* firm.

Public information performs three key roles in credit markets: (i) it reduces information asymmetries between the borrower and the lender, (ii) reduces information asymmetries between lenders and (iii) reduces a borrower's incentive to default. However, public information does have drawbacks. First, it may reduce a lender's incentive to procure information if the information will be later revealed by a credit registry. Second, public information may be imprecise or noisy causing excessive volatility in the observed public information (Morris and Shin (2002)).

Public information is often considered a substitute for private information, or a leveler of the playing field, but this is not always true. For example, consider a scenario where two banks can lend to the same firm but only one bank knows the identity of all the directors of this firm. Providing a directory of director creditworthiness to both banks, would only benefit the bank who is privately informed about the firm's composition of directors. Therefore, it is possible for public information to be a complement to private information.

Hertzberg et al., (2011) empirically demonstrate how merely the prospect of publicly announcing a bank's private credit rating about a firm can lead to strategic effects by lenders, and the subsequent reduction of credit. Lenders strategically reduced lending to borrowers who they had previously labeled as a poor credit. They argue that lenders reduced lending since the public information revelation would lead to other lenders reappraising their lending terms and potentially reducing their credit lines to the borrower, causing the borrower financial stress.

Our paper is similar to Hertzberg et al. in that we analyze the effects of altering public information in an environment with multiple lenders. Since the change in public information is common to all lenders, lenders must take into account how the change in public information affects their willingness to lend *and* how the change affects their competitors' willingness to lend.

Although both Hertzberg et al. and our paper examine changes in public information, the form of the public information is quite different. Hertzberg et al. analyze a reform which publicly released a lender's appraisal of a firm's creditworthiness, whereas in the reform that we analyze, the credit registry altered the information available about the firm's ongoing credit history. Hertzberg analyzes a reform where lenders respond to the expectation of a bad credit rating becoming public knowledge limiting the firm's capacity to procure credit; by contrast, we study a reform where the SBP removed the ongoing capacity to monitor a firm.

This paper contributes to the long literature on relationship banking (starting with Petersen and Rajan (1994); Berger and Udell (1995); Degryse and Van Cayseele (2000)). The prior literature has highlighted the potential for banking relationships to overcome the problem of information problems. Our paper emphasizes that the strength of these prior relationships does not completely eliminate the problem of imperfect information.

There is a nascent literature examining how variation in private information affects loan officers' decisions. Hertzberg et al., (2010); Cole et al., (2012); Paravisini and Schoar (2012) all examine how loan officers evaluate whether they should offer loans, and at what terms, under different information structures. This is similar to our paper's

environment, except the loan officers must consider not only what information they can observe but what other information may be available to other lenders.

Our paper examines the importance of interfirm relationships in a firm's ability to procure credit. Khwaja and Mian (2005a, 2008); Khwaja et al., (2011) show two key and related points: first, interfirm relationships allow insurance against idiosyncratic shocks; and second, politically connected firms are able to garner political favors for the entire set of related group firms. There is similar evidence in developed economies; Haselmann et al., (2018) show how social connections between banks and firms can facilitate more favorable lending terms for the firm. This paper emphasizes a mechanism for improved access to credit that is distinct from the intragroup lending channels studied in previous work. In particular, we suggest that the interfirm relationships may help firms procure loans since a lender considers the creditworthiness of both the firm and the firm's group. Therefore, being part of a group with a perfect credit history can facilitate a firm's access to credit. To our knowledge, this is the first paper that shows that information about a firm's group has implications on a firm's ability to borrow.

To summarize our paper has rich, intertemporal, administrative data and a unique natural experiment that ensures that our estimated effect for the change in public information on differentially informed lenders is well identified. Conversely, our paper analyzes a small set of business groups in Pakistan. As such, it is open question how similar results will be for other settings. For instance, the effect of changes in public information will likely be affected by differences in information transparency, and the quality of corporate governance.

3. Data and institutional background

This paper analyzes bank lending, which is the primary source of formal funding in Pakistan. For instance, in 2002, Pakistan's main stock exchange, the Karachi Stock Exchange, only had a market capitalization of 16 percent of GDP which is much smaller than the more mature NYSE which had a market capitalization of 92 percent of GDP (Khwaja and Mian (2005b)). The small size of public equity and debt issuance is a common feature amongst emerging stock markets. In part, this is due to institutional failings; for example Khwaja and Mian (2005b) show that brokers in Pakistan manipulated public stock prices through "pump and dump" schemes to earn rates of return 50 to 90 percentage points higher than outside investors.

The data comes from the SBP's electronic Credit Information Bureau (e-CIB), which legally requires all banks and lending institutions to submit data on all borrowing firms with outstanding loan amounts greater than 500,000 Pakistani Rupees (equivalent to about \$8500 in 2004).⁵ Some of the information collected by the SBP was passed back to the banks to facilitate lending. The information was provided through "credit worthiness reports." A sample report is shown in Fig. 9 in the appendix.

The creditworthiness report provided information about the firm's total borrowing, overdue loans, ongoing litigation against the firm, and amounts written off in the last five years. In addition, the central bank provided information on a firm's group borrowing, that is, the total borrowing by all firms which shared a mutual director, and whether the group had any amount overdue. The central bank would also report which group firms were overdue but the bank did *not* provide information on which firms were in the firm's group. Therefore, the individual banks would have different information about who—and who was not—in the firm's group.

⁵ This limit was removed in April 2006, and all loans (regardless of size) were required to be reported to the e-CIB.

Financial institutions use the credit reporting system as an initial appraisal and to monitor the ongoing creditworthiness of a firm. In an interview a former loan officer remarked: “The eCIB is used to verify credit history and monitor exposure, both during and after approval [of the loan].” Additionally, the banks’ written notes on a firm’s credit worthiness mention that credit reports were checked.

In May 2004, the central bank announced its intention to change the information in firm’s credit reports, but did not announce the implementation details or a detailed timeline. It took until April 2006 for the central bank to implement these new policies. Specifically, the central bank reduced the amount of information they would provide about a firm’s “group.” An annotated example of the new report is shown in Fig. 10 in the appendix. The main difference between the reports is that the key terms detailing the group’s outstanding loans were removed in 2006, hampering the lending institution’s ability to conduct due diligence on the firm and group.

Further there is important information in the creditworthiness of a firm’s group. As we show in more detail in Section 3.1, the probability of any loan to a group firm in our dataset being overdue in December 2004 was only 2.6 percent yet conditional on any firm in the firm’s group being overdue on a loan, that firm was significantly more likely to overdue. This highlights the importance of understanding the creditworthiness of the firm’s group in assessing the creditworthiness of the firm.

This paper argues that this reduction in information had a major impact on bank lending decisions. Specifically, it led banks to lend more to those firms about which they had private information.

3.1. Why did the State Bank of Pakistan alter the information available to lending institutions?

Prior to April 2006, the SBP defined a firm’s group as those firms which shared a director. However, this definition of groups was altered because the SBP believed that the definition of a group was too broad and not an adequate measure of control.

The following quote comes from the minutes of an interview with the head of the Banking and Regulation at the SBP—the group responsible for designing and implementing the prudential regulation titled “Criteria of Grouping Companies for the CIB Report,”—and outlines the motivations behind the change:

“The 2004 criteria of grouping for the purpose of CIB reports resulted in tying together some companies/firms/individuals, otherwise historically financially sound, with defaulters. This happened due to common directorship definition which also includes nominee directors who have little influence on the management of the firm. Hence, the SBP management decided to recognize and differentiate between those controlling shares in a group from common directors having no influence over the management of the group in question. Notwithstanding, SBP also appreciated financial institution’s need to take informative decisions if complete information of allied companies is not provided in the credit worthiness reports.”

These views were represented in the confidential internal minutes of a meeting of the SBP in May 2004. The State Bank’s intention was to provide a new definition for groups—one which would offer a better measure of control than the previous definition. However, the SBP delegated the responsibility for constructing the group relations to the banks, stating that “the onus for correct formation of the group as per definition given in the Prudential Regulations will be on Banks” (State Bank of Pakistan, 2004 Prudential Regulation). So the banks—the main beneficiaries of receiving information about a firm’s group—were the ones expected to provide the information to the SBP.

Moreover, the SBP specifically warned against reporting too large a group, stating that: “Banks are advised to be very careful while report-

ing the names of group entities in the CIB data. In case any party disputes the group relationship, the reporting Bank should be able to defend its position with documentary evidence” (State Bank of Pakistan, 2004 Prudential Regulation), further reducing the bank’s incentive to report a firm’s group to the SBP.

Ultimately, the regulation led banks to report almost no group information to the central bank. Since the SBP was no longer constructing group liabilities, and banks were not reporting group entities, banks were left to construct their own groups. Therefore, they were forced to make their own definition of groups and conduct their own due diligence. A current credit officer remarked: “With the new e-CIB system, you have to do your own intel and also consider past information for firms” and in the new system “the full group information was not declared—overall I preferred the older system.”

3.2. Other sources of firm information for a bank

Lenders access multiple different sources of information about a borrower prior to offering a loan. They collect information from the borrower using a standard form—the Basic Borrower Fact Sheet (BBFS), the firm’s accounts, and some information held by the Securities and Exchange Commission of Pakistan (SECP). Lenders also consult other people in the banking industry.⁶ The information reported in the BBFS and SECP detail, who are the firm’s directors and the amount of shareholding each director holds.

Thus, if a lender has a loan application from two firms who share a mutual director, using the information from the BBFS and the SECP, the lender can determine the existence of an interfirm relationship and the strength of the relationship (as proxied via the shareholdings). Further, we suspect if a bank lent to two members of the same group, the bank would have better information about that firm’s group.

3.3. Building firm links

The SBP collects information on all directors of a firm that borrows from a bank. This includes the director’s name, father’s name, a common identifier, shareholding, and home address. At the baseline of December 2004, we have details of 174,244 director relationships and 97,449 enterprises.⁷

3.4. What does the network of firms look like?

Fig. 1 shows the network of firm connections across the entire set of borrowing firms in Pakistan in December 2004. A connection between two firms is shown if both firms have at least one mutual director.

Only firms with at least one connection are shown in the graph above. The circles and edges in the picture correspond to firms and connections between firms respectively, where a connection is defined if both firms have at least one mutual director. Those firms with more connections are slightly darker and larger in the diagram.

The most visually striking aspect of the network is the huge dense network of firm connections in the center of the figure. The largest

⁶ It was surprising to us that loan officers mentioned talking to other banks to learn more about a client’s credit worthiness. A further way to procure knowledge of a firm’s group is via the SBP, lenders can query which firms a director is part of; however, our interviews with loan officers suggest that the knowledge of this facility is not apparent.

⁷ The e-CIB is missing the common identifier for a total of 19,473 of these director relationships and these are omitted from our sample. Since the common identifier was also used to compile which firms were in groups, our definition of which firms were in groups should not be affected.

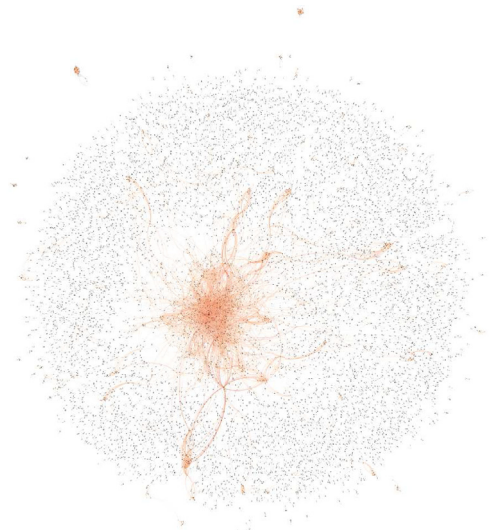


Fig. 1. The network of firm connections.

component⁸ is a total of 2,395 firms in December 2004.⁹

Fig. 12 in the online appendix shows what the network of firms looks like if we restrict attention to interfirm relationships where a mutual director owns at least 25 percent of the firm's equity. Fig. 3 shows the distribution of the number of connections each firm.

3.5. The data

The loan level data comes from the SBP's credit registry. The data includes information on all directors for each firm and the amount of equity each director holds. In December 2004, the data set included data on a total of 97,449 enterprises; 11,395 are corporate firms, and 86,053 are sole proprietorship or consumer loans. In April 2006, as part of an overhaul of the credit registry database, two separate registries were created: (1) corporate firm loans and (2) all consumer loans. The data on sole proprietor loans was moved into the consumer database and the data on corporate firms was kept within the corporate database. Simultaneously, some of the original corporate firms were placed in the consumer dataset—therefore, to maintain a consistent dataset of firms throughout the period, we use the set of firms that were ex-post defined to be corporate firms and maintained within the credit registry. Further, we exclude publicly listed firms due to our expectation that these firms should not be effected by the reform and a lack of sufficient sample size.

The SBP defined a firm's group as the set of firms with a common director in the entire borrowing data set of 97,449 enterprises. This paper follows the same definition of a group, using the entire database of 97,449 enterprises in December 2004. In total, our paper analyzes data on 3,695 group firms.

The data set stretches from December 2004 to December 2008; however there are data validation concerns immediately following the regulation change. The new credit registry required more information on borrowers to be uploaded by banks (for SBP's role of supervising

of the banking system, not for the purposes of the credit registry¹⁰) and subsequently required banks to upload the data in a new format which led to some initial teething problems. For example, loan records in the system were larger by a magnitude of ten for a single month, or there were seemingly duplicate loans, which only appear for one month. We use two methods to overcome these data validation concerns. First, in all our main within-firm regressions we include a bank interacted with time fixed effect; therefore, if there was any time-varying bank-specific measurement error (which was common across all the bank's loans), then this fixed effect should alleviate this concern. Second, in our baseline results, the months from April 2006 to August 2006 are dropped from the data set. Reassuringly, as a robustness test, including or excluding these months does not effect our results.

In addition, one of the banks, Union Bank, was taken over in late 2006. Since data reporting was poor during the takeover, that bank is omitted from all specifications.

The paper examines "funded" loan balances. A funded loan balance is either a term loan or a utilized overdraft facility. On the other hand, a non-funded loan balance is a bank's potential loan commitment to a firm, such as a bank's letter of credit.

The majority of lending are working capital balances, which are normally renewed every 12 months. They are similar to an overdraft facility, where firms are able to borrow more or less at any stage subject to their total borrowing limit. Unfortunately, data on the borrowing limit was not collected by the SBP prior to April 2006. Therefore, the paper restricts attention to the total amount borrowed in any month.

We use the official SBP definition for a nonperforming loan, specifically, a loan is defined if the loan amount due was overdue for more than 90 days.

There are a total of 94 banks which offer loans to corporate firms in 2004, but the sample is restricted to banks offering at least 50 corporate loans in December 2004. This leaves a final data set of 55 banks and 96.4 percent of the original data.

The sample is restricted to both harmonize the data set and for tractability. Many of the specifications use three high dimensional fixed effects: "firm interacted with date," "firm interacted with bank," and "bank interacted with date." Therefore, by removing the very smallest banks, the computation is greatly sped up.

All variables which include nominal amounts have been discounted to December 2004 prices using the official Pakistani CPI index published by the SBP.

Table 1 presents summary statistics for the group firms. Fig. 13 in the online appendix shows the distribution of total group firm borrowings and the distribution of loan sizes for group firms.

3.6. How informative was the credit report information about a firm's group?

As we detailed in Section 3.1, the SBP decided to change the credit report information following corporate lobbying and their own deliberations that the credit report information could be improved. Specifically, the definition used for a business group was too broad. However, even if the information was not ideal, it did provide banks with some important information about a firm's creditworthiness.

To show the importance of this information, we regress whether a firm's loan is overdue on whether a loan in the firm's group is also

⁸ A component is defined as, the set of nodes such that every node within the component has either a direct or indirect connection to every other node within the component.

⁹ It should be noted, we are only building the component of firms using the set of firms currently borrowing. Therefore, the giant component should really be seen as an underestimate of the true size of the largest component since there are possibly non borrowing firms which would link other firms into the giant component.

¹⁰ It is important that the extra information being collected in 2006, is solely for purpose of SBP's role of supervising the credit market since if the credit registry was displaying more information than it was in 2004, this could potentially conflate some of the observed effects of the regulation change.

Table 1
Summary statistics for group firms in December 2004.

Number of group firms	3695	
Number of loans	7250	
	Uninformed Loans	Informed Loans
Number of loans	5867	1383
	Mean	Mean
Overdue at baseline	2.77%	2.02%
Loan size (in '000s PKR)	56,343	49,791
Log loan size	9.28	9.43
Public banks	7.0%	7.6%
Private domestic commercial bank	56.5%	66.4%
Non-Bank Fin. Corp. (NBFC)	28.2%	17.3%
Foreign bank	8.3%	8.7%

A loan is defined at a firm-bank pair. Therefore, if a firm has multiple loans from the same lender, it is classified as a single loan. An informed loan is defined as a loan where at least two firms in the same group borrow from the same lender. In the following regressions, the paper shows how a firm that has both a informed and uninformed loan responds to the change in information.

Table 2
Is the performance of a group's loans correlated across firms (overlapping director)?

	Overdue	Overdue	Overdue	Overdue
Group firm overdue	0.0627*** (0.0137)	0.0608*** (0.0133)	0.0622*** (0.0133)	0.0667*** (0.0132)
Log loan size			−0.00439*** (0.00117)	0.00260* (0.00143)
Observations	7250	6569	6569	6569
Bank FEs	Yes	Yes	Yes	Yes
Business Sector FE	No	Yes	Yes	Yes
Province FE	No	Yes	Yes	Yes
Decile of firm borrowing	No	No	No	Yes

Standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The regressions in this table regress the performance of a firm's loan on whether another firm in that firm's group (where the group is defined as the set of firms that have at least one overlapping director) is overdue on their loan with additional controls, in December 2014. The regressions clearly show that conditional on at least one other firm in the group being overdue, this firm is more likely to also be overdue. Moreover, the regressions show that this correlation is robust to additional controls, such as controlling for the size of the loan, the business sector of the firm, the province of the firm, and the decile of the firm's total borrowings. All standard errors are clustered at the level of the component.

overdue, while controlling for firm characteristics. Specifically, we run the following regression:

$$\text{Loan overdue}_f = \beta \times \text{Group firm overdue}_f + \text{controls}_f + \epsilon_f$$

This regression tests whether a firm is more likely to have an overdue loan if another firm in the group is overdue on its loan, while controlling for various dimensions that may lead to a higher correlation. If β is positive and statistically significant, this would suggest that even after controlling for a firm's characteristics, the performance of the firm's group has useful information for the firm's creditworthiness.

In our regressions, we use all loans to group firms in December 2004, the first month of our dataset. To test the importance of varying the strength of firm's inter-relationships, we use two different definitions for a firm's group. In Table 2, we define the firm's group as the set of firms that have a common director with this firm; in Table 3, we define the firm's group as the set of firms that have common director and at least 25 percent shareholding in both firms, as of December 2004.

The probability of any loan to a group firm being overdue in December 2004 was 2.6 percent yet the results in Tables 2 and 3 show that conditional on any firm in the firm's group being overdue on a loan, that firm was significantly more likely to be overdue on its loan, even after controlling for the firm's characteristics. In Table 2, we show that if at least one other firm in the firm's group is overdue, that firm is 6–7

percentage points more likely to be overdue on their loan than if there are no other firms in the group that are overdue. This result is robust to adding various controls, such as controlling for the bank granting the loan, the business sector of the firm, the province of the firm, the loan size, and the firm's total borrowings.

In Table 3, we show this effect is greater for those firms with stronger director ties. Using a relatively stricter definition for a business group, we find relatively larger effects. Specifically, if at least one other firm in the firm's group is overdue, that firm is 18–20 percentage points more likely to be overdue on their loan than if there are no other firms in the group that are overdue. Similar to the results in Table (2), this result is robust to the inclusion of various other dimensions that could drive the correlation.

Moreover, Pakistani banks have historically poor recovery rates on nonperforming loans—emphasizing the need to conduct sufficient due diligence prior to offering a loan. For example, in 2005, Pakistani banks recouped a mere 14.1 percent of the value of the loans that were classified as nonperforming assets (Ministry of Finance (2006)).

Taken together, these results show the financial position of a firm is closely related to the financial position of the group and that this relationship is stronger when ties between the directors are stronger. Moreover, given the low recovery rates on nonperforming loans makes banks more cautious in lending.

Table 3

Is the performance of a group's loans correlated across firms (overlapping director with 25 percent shareholding)?

	Overdue	Overdue	Overdue	Overdue
Group firm overdue	0.196*** (0.0606)	0.180*** (0.0612)	0.180*** (0.0609)	0.180*** (0.0601)
Log loan size			−0.00336*** (0.00114)	0.00276* (0.00141)
Observations	7250	6569	6569	6569
Bank FEs	Yes	Yes	Yes	Yes
Business Sector FE	No	Yes	Yes	Yes
Province FE	No	Yes	Yes	Yes
Decile of firm borrowing	No	No	No	Yes

Standard errors in parentheses * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The regressions in this table regress the performance of a firm's loan on whether another firm in that firm's group (where the group is defined as the set of firms that have at least one overlapping director with at least 25 percent shareholding in both firms) is overdue on their loan with additional controls, in December 2014. The regressions clearly show that conditional on at least one other firm in the group being overdue, this firm is more likely to also be overdue. Moreover, the regressions show that this correlation is robust to additional controls, such as controlling for the size of the loan, the business sector of the firm, the province of the firm, and the decile of the firm's total borrowings. All standard errors are clustered at the level of the component.

4. The effect of the reform on a firm's source of credit

4.1. Econometric specification

The paper's main question is: What is the effect of asymmetric information between lenders on a firm's source of credit? We answer this question using a reform that exogenously reduced banks' information about firms in a way that varied across bank-firm lending pairs. We examine whether those banks with private information about a firm's group were more likely than other banks to renew a firm's loans following the reform.

In the paper's main specifications, a "group" is defined using a relatively narrow definition: An overlapping director must own a substantial amount of equity in two firms before those two firms are grouped together. We will demonstrate that altering the definition of a group in economically meaningful ways will lead to different results.

Definition 1. Firm f 's group is all other firms with whom firm f has a common director and at least 25% shareholding in both firms as of December 2004.¹¹ that groups are not mutually exclusive. Therefore, a group is defined with respect to a firm. For a quick pictorial representation how firms can be in multiple groups see Fig. 11 in the online appendix.

The main source of identification in the paper will be to compare borrowing amounts for the same firm from two different lenders, before and after the regulation, where each lender tends to different members in the group at baseline (so we are restricting the sample to lenders who were already lending before the reform, and therefore, by definition have some information). For example, assume there are two firms "F1" and "F2" who are in the same group. Both "F1" and "F2" borrow from two banks each in December 2004. "F1" borrows from banks "A" and "B" and "F2" borrows from banks "B" and "C," as shown in Fig. 2.

Notice that bank "B" is lending to both firms in the group in December 2004, whereas bank "A" and bank "C" only lend to one firm in the group. After the regulation change, only bank "B" is able to compile the group's lending.

Definition 2. A loan from bank b to firm f is labeled a "informed loan" if at least one other member of firm f 's group borrows from the same bank, b , in December 2004 (baseline). Similarly, a loan from bank b to firm f is

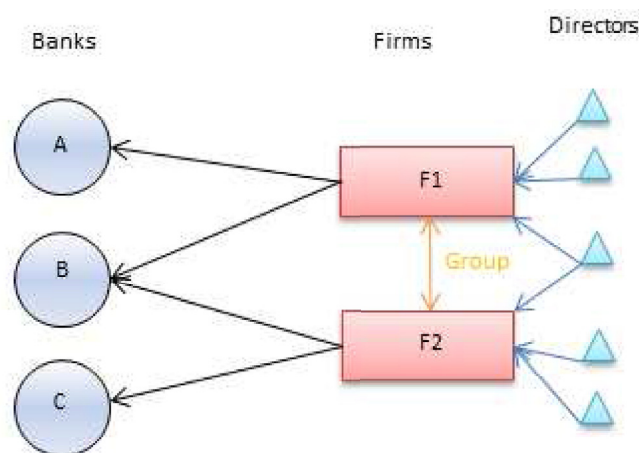


Fig. 2. Identification strategy: comparing loan details for the same firm borrowing from two different banks.

labeled an "uninformed loan" if no other member of firm f 's group borrows from the same bank, b , in December 2004.

In the stylized example in Fig. 2, firm F1 has one informed loan (the loan with bank "B") and one uninformed loan (the loan with bank "A"). The paper's identification will be examining how lending changes between the informed and uninformed loans, before and after the regulation change. Fig. 4 shows the distribution of loan sizes between informed and uninformed loans are relatively similar.

We estimate equations of the form:

$$Y_{bft} = a_{bf} + a_{bt} + a_{ft} + \beta_1 \times \text{post}_t \times \text{Informed}_{bft} + \epsilon_{bft}$$

The unit of observation is at the bank-firm-date level, so Y_{bft} is the variable of interest at bank b , firm f in month t . For example, it could be the size of the loan outstanding by firm f at bank b , in month t . Informed_{bft} is a dummy variable equal to one if the loan between bank b and firm f is an informed loan. Post is a dummy variable equal to one for a loan after April 2006.

The standard errors ϵ_{bft} are clustered at the level of the component—at the level such that every firm within the component

¹¹ This definition of groups allows firms to be in multiple groups and that groups are not mutually exclusive. Therefore, a group is defined with respect to a firm.

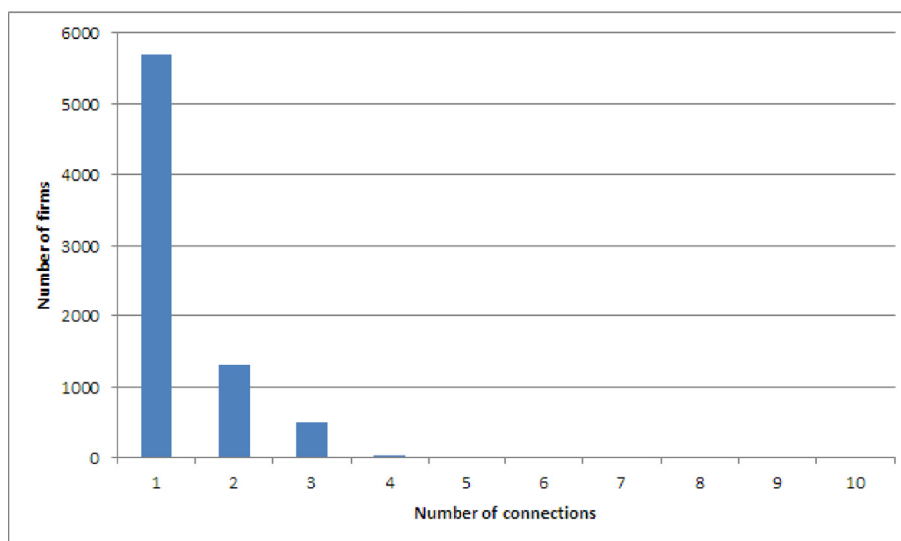


Fig. 3. The number of connections for each firm. This figure displays the number of connections each firm possesses in December 2004 where we define two firms to be linked if they share a mutual director who owns 25 percent or more of the firm's equity.

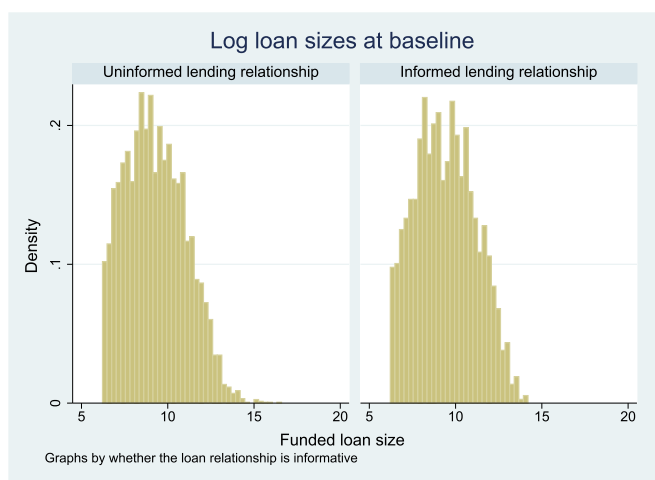


Fig. 4. The histogram of log loan size for informed and uninformed banking relationships. This is the histogram of the log loan size in December 2004 separated by informed and uninformed banking relationships. Since, no loan below 500,000 PKR was reported to the credit registry, there is a sharp cut-off at 6.2 (approximately the log of 500). We provide robustness tests to ensure the cutoff of small loans is not driving in our results.

has at least one indirect link to every other firm in the component.¹²

All the main regressions contain a firm $\times$ date fixed effect, α_{ft} . This fixed effect implies that we are estimating the difference in firm lending using differences for the same firm and in the same month. Therefore, we are estimating β_1 from only those firms that have both an informed and an uninformed loan. We have a total of 449 firms and 1784 loans in December 2004 who had both an informed and an uninformed lending relationship. Table 4 presents summary statistics about the firms that identify β_1 .

¹² A component is defined as the set of firms for whom there exists a direct or indirect link to every other firm within the component. A *direct link* between two firms is defined if there is at least one mutual director who owns at least 25% of each firm and an *indirect link* exists between two firms if there is at least one *path* of direct links between the two firms. Conceptually, if firm f is in component c then firm f 's group must be a subset of a component c . The definition of a link, is similar to the one used in all benchmark specifications.

Those group firms that have both an informed and uninformed loan account for just over 12 percent of group firms that were borrowing in Pakistan in December 2004. Admittedly, the number of firms is relatively small, but the total amount borrowed by these group firms was substantial. The total loans to the group firms that had both an informed and uninformed loan in December 2004 was over 96 billion PKR (over 1.5 billion USD), over 10 percent of all total corporate borrowing, and over 24 percent of all group firm borrowing, in 2004. Therefore, these group firms were relatively large and a significant component of the Pakistani economy suggesting that changes in credit access to these firms could have material effects.

Fig. 5 demonstrates that the firms who identify β_1 in general have larger loans. This is in part mechanical, since we are only identifying the effect of the policy from those firms who have at least two loans.

Including the fixed effect a_{bt} ensures that we are allowing for any aggregate change in bank lending for each month, and the fixed effect a_{bf} ensures we are controlling for any firm-bank specificity.

This section restricts attention to those banks that were lending in December 2004; therefore, the paper does not include any new relationship in the analysis because any new borrowing relationship may be endogenous. Furthermore, in the estimation procedure, firms who discontinue relationships with all of their original lenders are dropped.¹³

4.1.1. Outcomes of interest

There are three main outcomes of interest in the paper:

- Log loan size $_{bft}$
- Renewed loan $_{bft}$
- Overdue $_{bft}$

"Log loan size $_{bft}$ " is defined as the log of real funded loan size outstanding in date t at bank b by firm f . If there is no loan observed this is coded in the data as equal to the minimum of what is observable in the data set (the log of 500,000 PKR).

¹³ Out of the original 3,695 group firms that were borrowing in December 2004, there were 2,702 group firms with at least one loan in December 2017 (three years after the start of our sample period) and there were 2,332 group firms with at least one loan in the last month of dataset (December 2018). To ensure the attrition of firms from our dataset does not drive our results the paper also shows robustness results showing similar sized effects with the set of firm-bank connections that always maintain an active borrowing relationship.

Table 4

Summary statistics for group firms who have both an informed and uninformed loan in December 2004.

Number of group firms	449	
Number of loans	1784	
	Uninformed Loans	Informed Loans
Number of loans	1035	749
	Mean	Mean
Overdue at baseline	1.55%	1.33%
Loan size (in '000s PKR)	45,667	65,162
Log loan size	9.18	9.55
Public banks	5.4%	5.0%
Private domestic commercial bank	48.9%	63.0%
Non-Bank Fin. Corp. (NBFC)	37.9%	23.1%
Foreign bank	7.3%	8.9%

A loan is defined at a firm-bank pair. Therefore, if a firm has multiple loans from the same lender, it is classified as a single loan. Since the estimation results in section 4 are estimated from firms who have both an informed and uninformed lending relationship, this is the set of firms which identify our main coefficient of interest.

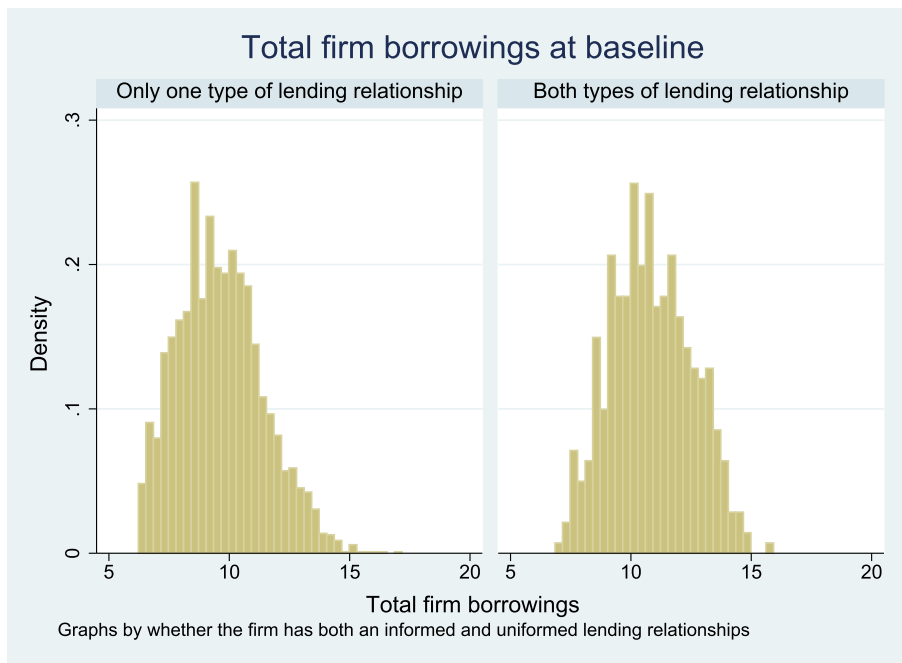


Fig. 5. The histogram of total firm borrowings by lending relationship. This is the histogram of total firm borrowing in December 2004 separated by the firms we are using for identification in section 4. We are only identifying the effects of the policy from the set of firms who borrow from both an informed and uninformed lender, therefore, the right panel demonstrates the distribution of total firm borrowings who have both an informed and uninformed lending relationship. As you would expect the distribution of total borrowings for firms who have both types of relationship are larger. This is partly mechanical effect as they must have at least two loans. The larger total borrowings at baseline would suggest these firms are also larger in general.

“Renewed loan_{bft}” is defined as whether firm f at bank b at date t has an outstanding funded loan amount above 500,000 PKR (in 2004 Pakistani rupees).

“Overdue_{bft}” is defined as whether firm f at bank b at date t is overdue at date t . There are certain endogeneity issues when looking at overdue rates because a firm can only go overdue if a firm has a loan which will be shown to be a function of the amount of information a bank has about a borrower. This is discussed in greater detail in the results.

4.2. Results

We explore the implications of differentially informed lenders on a firm's source of credit in three ways: (i) to explore the overall effects of the reform, we estimate the difference in loan sizes between informed and uninformed lenders; (ii) to explore the role of private information, we estimate how the difference in loan sizes changes as we use various measures of informed lenders; and (iii) to explore whether there were heterogeneous effects for different firms, we estimate the effect of the reform on different firm sizes and differing credit risks.

4.2.1. Following the reform, loan sizes were relatively larger within informed banking relationships

Our main results indicate that when public information available to banks was reduced, the change caused those banks to lend more to firms for whom they had greater private information. Banks with a better knowledge of a firm's group were more likely to renew the credit facility, and grant greater credit for renewed loans.

This is most strikingly represented in Fig. 6. Fig. 6 plots the coefficients from a regression of loan renewal on an informed dummy, interacted with date dummies, and all second-order fixed effects, firm $\times$ date, date $\times$ bank, and firm $\times$ bank. The figure clearly shows that the difference in renewal rates between informed and uninformed loans is relatively constant before the reform, suggesting that the common trends identifying assumption holds. Following the reform, there is a sharp and persistent increase in the renewals of informed loans relative to uninformed loans. This indicates that the reform causes banks to increase their lending to firms for which they have more information.

Table 5 shows the estimates for the policy's effect under different specifications. The first column contains date, firm and bank fixed

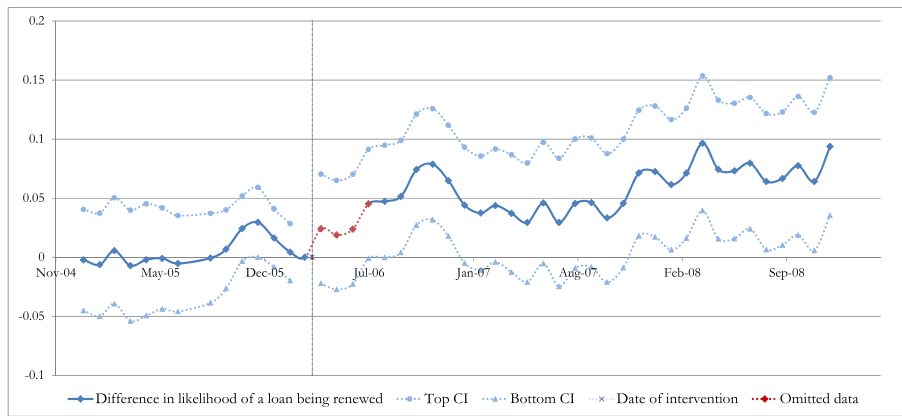


Fig. 6. The monthly coefficients for the difference in likelihood of a loan being renewed. This is the monthly coefficient for *informed* loans, when using ‘firm × date’, ‘firm × bank’ and ‘bank × date’ fixed effects. The light blue lines are point-wise 95% confidence intervals. The points in red (and labeled “omitted data”) are the months with data validation concerns (as outlined in section 3.5). For transparency the coefficients for these months are shown in this figure, and it is visually clear they do not change the general trend that the informed lending relationship was more likely to have their loan renewed. This figure visually demonstrates there was a dramatic increase in the likelihood of a loan not being renewed for the same firm when the borrowing relationship was an uninformed banking relationship. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 5

The effect of the policy on the likelihood of a loan being renewed between informed and uninformed lending relationships.

	(1)	(2)	(3)	(4)	(5)	(6)
Informed	0.00213 (0.0150)	0.0213 (0.0135)	−0.00830 (0.0135)		0.00453 (0.0126)	
Post*Informed	0.0562*** (0.0121)	0.0734*** (0.0244)	0.0722*** (0.0241)	0.0767*** (0.0242)	0.0517** (0.0217)	0.0543** (0.0218)
Observations	269,625	269,625	269,625	269,625	269,625	269,625
Firm&Date FE	Yes	N/A	N/A	N/A	N/A	N/A
Date*Firm FE	No	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	No	Yes	N/A	N/A	N/A
Firm*Bank FE	No	No	No	Yes	No	Yes
Bank*Date FE	No	No	No	No	Yes	Yes

Standard errors in parentheses *p < 0.1, **p < 0.05, ***p < 0.01.

Each regression shows the likelihood of an informed loan to be renewed to be 5 to 7 percentage points more likely. All the specifications 2–6 include a date × firm fixed effect, therefore, we identify the likelihood of a informed loan to be renewed solely from the set of borrowers who have both a informed and uninformed loan. All standard errors are clustered at the level of the component.

effects, and therefore is being estimated from between- and within-firm differences. Columns 2 through 6 all include a firm interacted with date fixed effects. This ensures that we only use the set of borrowers who have both, an informed and uninformed banking relationship, to identify whether an informed banking relationship was more likely to be renewed.

The estimates are all relatively similar and the various specifications suggest that an informed banking relationship was between 5 and 8 percentage points more likely to be renewed than an uninformed banking relationship for the *same* firm. The preferred specification is column 6, which includes all 3 s order-fixed effects. In doing so, we are controlling for any aggregate changes in a bank’s willingness to offer credit over time and firm-bank match specificity, and identifying the effect from firms which had both an informed and uninformed lending relationship. Fig. 6 is the graphical counterpart of the regression in column 6, except that we interact date dummies with the informed dummy variable.

While the previous results examine the effects of the policy on the extensive margin, here we can examine if the total size of the funded loan was larger in an informed banking relationship after the regulation change.

Fig. 7 plots the coefficients from the log loan size on an informed dummy variable interacted with date dummies and all second-order fixed effects.¹⁴ Though the estimates are less precise than the extensive

margin, we clearly see a similar trend as the extensive margin. Table 6 shows the same specifications as Table 5 but the dependent variable is the log of real funded loan size. The various estimates suggest that an informed banking relationship was between 8 and 12 percent larger following the policy change.

As a robustness check, we restrict the sample to those lending relationships which last until June 2008 or December 2008 and compare loan balances across informed and uninformed banking relationships for the same firm. Table 16 in the online appendix shows the estimates from this regression. Although power is an issue, the size of the effects look similar across the different specifications to the results in Table 6.

The SBP only required banks to report details of a loan if the firm’s outstanding loan was greater than 500,000 rupees (\$8500) and this could lead to the following bias in our results. Loans which were initially just above the cutoff could be partially repaid, subsequently falling below the 500,000 rupees threshold and as such incorrectly categorized as a non-renewed loan. This is a greater concern since the uninformed loans are in general smaller than the informed loans. Table 17 in the online appendix excludes firms which had a loan close to the cutoff in December 2004. The observed effect of the policy seems consistent when we exclude firms with a loans approximately below \$12,750, \$17,000, or \$21,250 in December 2004. This suggests that the censoring of the data at 500,000 rupees does not affect the results.

4.2.2. There were substantial differences in the effect of the policy on the credit market, depending on the strength of the interfirm relationship

The results in section 4.2.1 demonstrate that the reform led banks with private information about a group to lend more than other banks.

¹⁴ If a loan is not observed, we code the loan size to be the log of 500,000 PKR, which is the minimum threshold at which we are able to observe a loan.

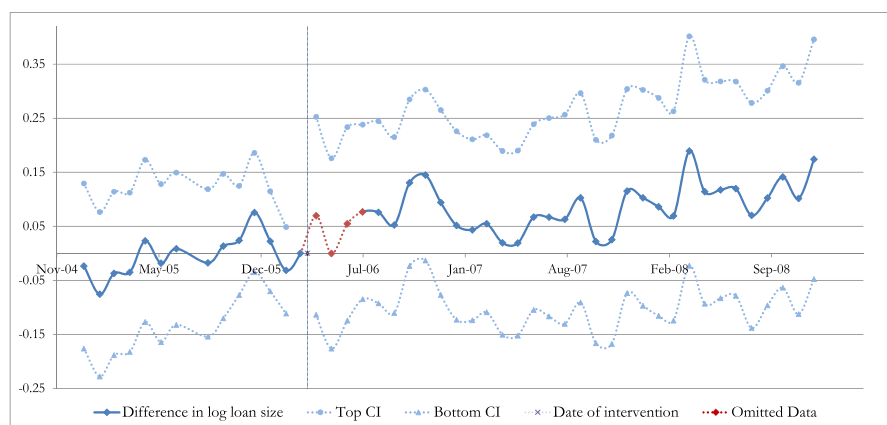


Fig. 7. The monthly coefficients for the difference in log loan size. This is the monthly coefficient for *informed* loans, when using ‘firm × date’, ‘firm × bank’ and ‘bank × date’ fixed effects. The light blue lines are point-wise 95% confidence intervals. The points in red (and labeled “omitted data”) are the months with data validation concerns (as outlined in section 3.5). For transparency the coefficients for these months are shown in this figure, and it is visually clear they do not change the general trend. There does not seem to be a pre-trend in the difference in log loan sizes but there is some evidence that the loan sizes increased after the regulation change. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 6

The effect of the policy on the loan sizes between informed and uninformed lending relationships.

	Loan Size	Loan Size	Loan Size	Loan Size	Loan Size	Loan Size
Informed	0.167** (0.0761)	0.449*** (0.0944)	0.216*** (0.0728)		0.227* (0.0709)	
Post*Informed	0.196*** (0.0480)	0.111 (0.0869)	0.120 (0.0854)	0.125 (0.0830)	0.0954 (0.0786)	0.0916 (0.0775)
Observations	269,625	269,625	269,625	269,625	269,625	269,625
Firm&Date FE	Yes	N/A	N/A	N/A	N/A	N/A
Date*Firm FE	No	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	No	Yes	N/A	N/A	N/A
Firm*Bank FE	No	No	No	Yes	No	Yes
Bank*Date FE	No	No	No	No	Yes	Yes

Standard errors in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Each regression shows the likelihood of a informed loan to be to be 9 to 13 percent larger following the reform. All the specifications 2–6 include a date × firm fixed effect, therefore, we identify the difference in loan size solely from the set of borrowers who have both a informed and uninformed loan. All standard errors are clustered at the level of the component.

Here we consider whether the effects vary by the amount of private information an informed lender has. In particular we consider variation in: the strength of the observed interfirm relationship and the number of firms who borrow from the same lender.

We observe substantial heterogeneity in the measured effect of the reform depending on the level of control implied within the interfirm relationship. If we separate the interfirm relationships according to the amount of equity a director owns, it is clear there was little or no effect on those firms which had only overlapping directors. Further, the size of the effect was increasing in the interfirm relationship according to equity levels, which is a proxy for control. Therefore, this suggests that merely overlapping directors had no informational content on a firm’s creditworthiness.

Table 7 and Fig. 8 show the effect of the reform for different levels of equity held by a mutual director. If a bank observed two firms who shared a director who owned at least 40 percent of both firms, the bank was 7.6 percentage points more likely to renew a loan from either firm, compared to a bank who only observed one firm. However if a director had no ownership stake in either firm, the bank was only 2.5 percentage points (not statistically significant) more likely to renew their loans.

4.2.3. The effect of the change in information was predominantly felt by small to medium sized firms

From a policy perspective it is important to examine what type of firms were most affected by the reform. We examine if the effect varies by firm size.

Table 8 and Fig. 14 (in the online appendix) show the effects of the policy on the likelihood of a loan being renewed by different deciles, where the deciles are created according to the total amount the firm borrows at baseline. We see almost no effect (and certainly no statisti-

Table 7

The effect of greater shareholding on the effect of the policy on loan sizes and likelihood of renewing the loan.

	Loan Size	Loan Renewed
Post*Informed 0%	0.0397 (0.0716)	0.0248 (0.0202)
Post*Informed 10%	0.0980 (0.0936)	0.0440* (0.0234)
Post*Informed 25%	0.170* (0.0993)	0.0555** (0.0275)
Post*Informed 40%	0.0218 (0.122)	0.0767** (0.0328)
Observations	269,625	269,625
Date*Firm FE	Yes	Yes
Firm*Bank FE	Yes	Yes
Bank*Date FE	Yes	Yes

Standard errors in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Construction of the ‘informed 40%’ variable is the same as the procedure in the paper for constructing the benchmark bank-firm pairs, except using a 40% cutoff for directorial shareholding. We construct the ‘informed 25%’ variable to be those firm-bank pairs who would be defined as informed using the usual definition with a 25% director shareholder cutoff - except omitting those firm-bank pairs that were already labeled as ‘informed 40%’. A similar procedure is used to construct the 10% and 0% informed variables. All standard errors are clustered at the level of the component.

cally significant effect) on the largest decile of borrowers.

These results suggest that public information was most important for small to medium sized firms. These results can be interpreted in two ways: (i) the relative cost of procuring information is largest for

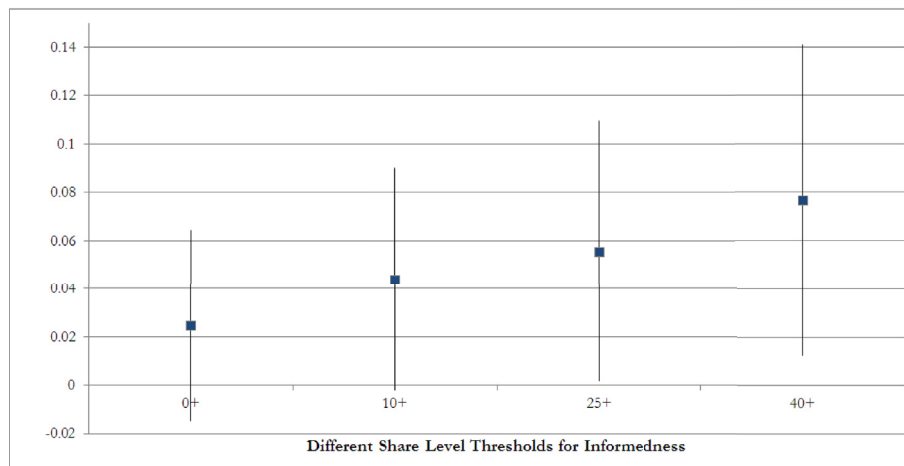


Fig. 8. The effect of the policy on the likelihood of the loan being renewed by different share thresholds. This figure shows the effect of the policy according to different definitions of informativeness. In particular, we create new dummies according to the interfirm relationship a bank observes. Those relationships where the bank observes two firms with solely overlapping directors has little to no effect on whether the loan is renewed. Whereas if a lender observes two firms with an interfirm relationship where there is a mutual director who owns more than 40 percent of the company, those firms' loans are 7.6 percentage points more likely to be renewed than if the lender only observed one of those firms. This is the visual analogue of column 2 in Table 7. The 95% confidence interval is depicted with the straight lines, and the estimated coefficient is the small blue box. All standard errors are clustered at the level of the component. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 8
The effect of the policy by decile of firm size.

	Loan Size	Loan Renewed
Post*Informed*2nd Decile by Firm Size	0.461** (0.232)	0.543*** (0.127)
Post*Informed*3rd Decile by Firm Size	0.247 (0.242)	0.166 (0.108)
Post*Informed*4th Decile by Firm Size	−0.0207 (0.207)	−0.122 (0.122)
Post*Informed*5th Decile by Firm Size	0.112 (0.212)	0.0607 (0.0783)
Post*Informed*6th Decile by Firm Size	0.356 (0.211)	0.109 (0.0738)
Post*Informed*7th Decile by Firm Size	0.0125 (0.215)	0.0262 (0.0686)
Post*Informed*8th Decile by Firm Size	0.134 (0.146)	0.158*** (0.0519)
Post*Informed*9th Decile by Firm Size	0.102 (0.174)	0.0225 (0.0446)
Post*Informed*10th Decile by Firm Size	−0.00955 (0.152)	−0.00843 (0.0355)
Observations	269,625	269,625
Date*Firm FE	Yes	Yes
Firm*Bank FE	Yes	Yes
Bank*Date FE	Yes	Yes

Standard errors in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each decile is constructed by computing the total firm borrowings in December 2004. The effect on the smallest decile is not identified in the data since the smallest borrowers in December 2004 only had one loan and the identification relies on a borrower having at least two loans. All standard errors are clustered at the level of the component. The results for column 2 are shown in Fig. 14.

the smallest firms or (ii) information about the largest firm's groups is already well known by the banks. We consider each possibility in turn.

As discussed in section 3.2, banks procure more information about a firm in addition to what is provided by the SBP's credit registry service. This information—such as calling other bank managers, or accessing the SECP database—may be costly to acquire. It is plausible that loan officers will conduct greater scrutiny over larger loans (assuming the cost of default is linear in loan size). Consequently the effect of the regulation change would be largest on those firms that borrowed the least (as this is where the proportional cost of acquiring more information to dollar lent is the largest).

A former Director of Banking Policy and Regulation Department at the SBP remarked in an interview: "The number of corporate [firms] is not large in Pakistan therefore people tend to know about reputations," further strengthening the assertion that we would expect little to no

Table 9
Firms who were overdue at baseline were more likely to renew their informed loan.

	Loan Size	Loan Renewed
Post*Informed	0.0681 (0.0778)	0.0477** (0.0222)
Post*Informed*Firm Overdue	0.472* (0.278)	0.133* (0.0691)
Observations	269,625	269,625
Date*Firm FE	Yes	Yes
Firm*Bank FE	Yes	Yes
Bank*Date FE	Yes	Yes

Standard errors in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. This is suggestive evidence that those firms who had been overdue in the past were more likely to renew their loans at an informed banking relationships. This suggests that those firms that are likely classified as the most risky were also the ones most likely to have their loans renewed by an informed lender. All standard errors are clustered at the level of the component.

effect on those firms that had large groups.¹⁵

4.2.4. The effect of the policy was largest on those firms who were overdue on a loan in December 2004

Another way to analyze how the reform may have had heterogeneous effects across firms is to consider those firms who were observably worse credit risks prior to the reform. Firms that had loans overdue in December 2004 are expected to be at the greatest risk of being overdue in the future. Further, the problem of incomplete information on the firm's group may be more pronounced, since one firm with an amount overdue might indicate financial distress within the wider group. So, a lending institution may be more willing to lend to a firm if it is able to inspect the wider set of group firms too. Also, when one firm defaults, the institution could react by reducing lending to the group at large.

Therefore, a firm that has an amount overdue may be more likely to borrow more from an informed banking relationship after the regulation change. Confirming this economic intuition, these results are shown in Table 9.

¹⁵ Table 18 and Fig. 18 (both in the online appendix) show the effects of the policy on the likelihood of a loan being renewed by different deciles, where the deciles are created according to the total amount the firm's group borrows at baseline. Similar to the results in Table 8, we see no effect on those firms who are part of the largest groups.

Table 10
The difference in the likelihood of a loan going overdue.

	Overdue	Overdue
Informed	0.0163*** (0.00518)	
Post*Informed	−0.0287*** (0.00666)	−0.00719 (0.0141)
Observations	209,962	209,962
Firm&Date FE	Yes	N/A
Date*Firm FE	No	Yes
Bank FE	Yes	N/A
Firm*Bank FE	No	Yes
Bank*Date FE	No	Yes

Standard errors in parentheses *p < 0.1, **p < 0.05, ***p < 0.01.

Caution should be taken when taking inference from these regressions. As described in the text, this is not estimating the causal impact of the regulation change on the likelihood of a loan becoming overdue, since in Table 5, we showed uninformed loans were more likely to be not renewed. Column 1 does not include firm interacted with date fixed effects. Therefore, column 1 suggests the loans that were renewed by uninformed lenders were more likely to default. Column 2 which includes the firm interacted with date fixed effect suggests that those firms for whom the informed and uninformed lender renewed, there was no differential in default. All standard errors are clustered at the level of the component.

4.2.5. Overdue rates are similar across informed and uninformed banking relationships

The previous results highlighted how the reform affected who was able to borrow. In this section we examine if the reform affected overdue rates.

The reduction in public information may alter a firm's incentive to default on a loan for two reasons. First, a firm may be more willing to be overdue because the impact on the rest of the firm's group will be limited. Second, the lender would be less willing to offer a loan. In section 4.2.1 we demonstrated that the uninformed lender was 5.4 percentage points less likely to renew a firm's loan than the informed lender.

Only if all loans were renewed would we be able to identify the effect of the change information on a firm's incentive to default.

However, if strategic default was a key problem in these corporate markets you would expect the relative likelihood of a loan being overdue to be greater at an uninformed lender than an informed lender for the same firm. The firm is less likely to have its loan renewed by the uninformed lender, reducing the dynamic incentive to repay the loan. Also, the repercussions on a firm's group would be smaller at an uninformed lender. Therefore, the absence of any differential on the likelihood of a loan being overdue between an informed and uninformed lender suggests there was no greater strategic default by the firm upon the uninformed lender.

The regressions in column 2 of Table 10, show little difference in overdue rates between informed and uninformed banking relationships. Column 2, which estimates the difference in the likelihood of a loan being overdue at an informed and uninformed lender for the same firm, shows a relatively precise zero estimate. Column 2 suggests there was no strategic default by the firm following the regulation reform—even though they were less likely to have their loan renewed.

Column 1 is estimated from both, between-firm and within-firm differences, since it does not include a firm × date fixed effect. Column 1 suggests uninformed lenders select a worse set of loans to renew than informed lenders. In particular, defaults rates for uninformed lenders were 3 percentage points higher than informed lenders following the reform.

5. The effect of the reform on a Firm's access to total credit

In the previous section we established that a firm was more likely to receive a loan from an informed lender following the reform. In this

section, we examine whether those firms with an informed lender were more likely to have larger total credit lines following the reform. Consequently, we ask: Did firms merely substitute their lending partners and receive the same total loan amounts? Or, did the reform lead to real effects in how much a firm was able to borrow?

5.1. Econometric specification

Definition 3. Firm f has an informed lending relationship if firm f borrows from a bank who also lends to at least one other member of firm f 's group in December 2004.

To examine whether the reform affected a firm's access to credit we create a dummy variable “informed lender” which takes a value of “1” if the firm had an informed lending relationship in December 2004.

We identify the effect of the reform by comparing total loan amounts between those firms with an informed lending relationship and those firms without an informed lending relationship, before and after the regulation. Formally, we estimate an equation of the form:

$$Y_{ft} = \alpha_f + \alpha_t + \beta_1 \times post_t \times \text{Informed Lender}_f + \gamma X_{ft} + \epsilon_{ft}$$

Where Y_{ft} is the log of real total funded borrowings by firm f in month t . Informed lender $_f$ is a dummy variable equal to one, if the firm has at least one informed lending relationship in December 2004. Post is a dummy variable equal to one for a loan after April 2006. α_f and α_t are firm and date fixed effects respectively. In section 4 our identification relied on comparing the loan outcomes for the same firm, whereas in this section we rely on comparing the total credit borrowed between firms. Consequently, to allow for firm differences across regions and sectors, we include a set of firm by time controls X_{ft} , such as the firm's province interacted with time fixed effects.

As before, the standard errors ϵ_{ft} are clustered at the level of the component—at the level such that every firm within the component has at least one indirect link to every other firm in the component. As in the previous section, we do not observe loans smaller than 500,000 PKR. Therefore, if we do not observe any loan for a firm, we code that firm's total borrowings to be 500,000 PKR.

5.2. Results

5.2.1. Firms with an informed lending relationship borrowed relatively more after the reform than those firms without an informed lending relationship

Table 11 estimates the effect of the reform on firms who had an informed lending relationship in December 2004. The difference-in-difference estimates suggest that those firms with an informed lender were able to borrow 11 to 14 percent more following the reform, compared to firms with no informed lender. The results are robust to various controls such as a province interacted with date fixed effect and business sector interacted with date fixed effect.

Combining the results from section 4.2.1 with the results in Table 11 suggests that the reform had two key effects: (i) lenders with greater information lent more to the same firm following the reform and (ii) those firms with informed lenders were able to access more credit following the reform.

This suggests that the reform not only led to a reallocation of lending across lenders but also led to real effects in the credit market by altering a firm's capacity to procure credit. Interpreting these results, the reduction in public information led to a reallocation of credit to borrowers for whom the lender had greater information. Ultimately, this disadvantaged firms who borrowed from banks which had limited information about their wider set of interfirm relationships.

Table 11

Difference in total loan sizes between firms who have an informed lender and those who do not.

	Total Loans	Total Loans	Total Loans	Total Loans
Post*Informed lender	0.130*** (0.0447)	0.134*** (0.0458)	0.113** (0.0464)	0.123** (0.0476)
Observations	161,744	155,496	149,248	145,684
Firm FE	Yes	Yes	Yes	Yes
Date FE	Yes	N/A	N/A	N/A
Group Size Specific Time Trend	Yes	Yes	Yes	Yes
Business Sector FE*Date FE	No	Yes	No	Yes
Province FE*Date FE	No	No	Yes	Yes

Standard errors in parentheses *p < 0.10, **p < 0.05, ***p < 0.01.

The difference-in-difference estimates suggest that those firms with an informed lender were able to borrow 11–14% more following the reform than those firms with no informed lender. Column 1 shows the estimates with firm fixed effects and a group size specific time trend. Column 2 shows the estimates when including business sector interacted with date fixed effects. Column 3 shows the estimates when including province interacted with date fixed effects. Lastly column 4 shows the estimates where we include all interacted fixed effects. We are missing the data on the province and business sector for a small fraction of the firms, therefore, they are omitted from those regressions which include province or business sector fixed effects. All standard errors are clustered at the level of the component.

5.2.2. The borrowing capacity of the smallest firms were most affected by the reform. The effect of the reform was greatest on the smallest firms' total borrowings

We have established that the reform had real effects on how much a firm was able to borrow. To further evaluate the welfare and policy implications, we ask: What type of firms were the most affected by the reform?

To answer this question we estimate the effect of the reform on different sized firms.

Table 12 estimates the effect of the reform on different deciles of firm size and whether they had an informed lending relationship in December 2004. The difference-in-difference estimates suggest that those firms in the smallest decile were the most affected by the reform.

Combining the results from section 4.2.3 suggests that public information was crucial for the smallest firms. Not only were the more informed lenders more likely to lend to the smallest firms, but the smallest firms with an informed loan were more likely to procure larger total borrowings. The change in total borrowing for the largest decile was not significantly different from zero. These results further support the evidence that the regulatory reform had significantly different effects for different sized firms, with the largest effects for the smallest firms.

6. Conclusion

Asymmetric information is a focal issue when studying credit markets. These information asymmetries may exist between a borrower and a lender, and between different lenders. To investigate the implications of adverse selection among differentially informed lenders, we use a reform by the State Bank of Pakistan (SBP).

In April 2006, the SBP instituted a credit registry reform which exogenously reduced public information about a firm's creditworthiness, and did so in a way that varied across firm-bank pairs.

We utilize the natural experiment that the SBP's reform generated. We show empirically that the absence of public disclosure of group affiliation led to a reallocation of corporate borrowing. Those lenders that had greater private information were 5.4 percentage points more likely to renew a borrower's loan. Remarkably, this is true even for banks that had a pre-existing relationship with the firm, suggesting that the strength of prior relationships does not eliminate the problem of imperfect information.

Not only do we see a reallocation of corporate borrowing, we also demonstrate that those firms who borrowed from an informed lender were able to borrow 12.3 percent more than firms who borrowed from less informed lenders. Consequently, we show that the source of a firm's

Table 12

The difference in total loan borrowings by firm decile and whether they had an informed lending relationship.

	Total Loans	Total Loans
Post*Informed*1st Decile	0.764*** (0.118)	0.756*** (0.137)
Post*Informed*2nd Decile	0.253** (0.103)	0.290** (0.121)
Post*Informed*3rd Decile	0.310*** (0.0957)	0.318*** (0.101)
Post*Informed*4th Decile	−0.274*** (0.104)	−0.311*** (0.119)
Post*Informed*5th Decile	0.217** (0.102)	0.173 (0.111)
Post*Informed*6th Decile	0.0678 (0.0988)	0.0708 (0.101)
Post*Informed*7th Decile	−0.0511 (0.129)	−0.0282 (0.128)
Post*Informed*8th Decile	−0.0714 (0.134)	−0.100 (0.136)
Post*Informed*9th Decile	0.267** (0.105)	0.257** (0.105)
Post*Informed*10th Decile	0.0340 (0.132)	0.0728 (0.143)
Observations	161,744	145,684
Firm FE	Yes	Yes
Date FE	Yes	N/A
Group Size Specific Time Trend	Yes	Yes
Business Sector*Date FE	No	Yes
Province*Date FE	No	Yes

Standard errors in parentheses *p < 0.10, **p < 0.05, ***p < 0.01.

These regressions interact a dummy variable for each decile (by total firm borrowings in December 2004) with the post × informed lender variable. Hence, we are estimating the effect of the policy separately for each decile. The estimates suggest that the largest effects were on the smallest firms, which is similar to the results shown in section 4.2.3. All standard errors are clustered at the level of the component.

credit, as well as the total credit it could procure, were affected by the SBP's reform.

Our paper contributes to the wider literature on the importance of information using a small set of business groups in Pakistan. As such, it is open question how similar results will be for other settings. For instance, are our results due to the lack of transparency in Pakistani group firms, or a symptom of poor corporate governance? Are banks worried about rent appropriation by related companies (Bertrand et al., 2002), or are they worried about the possible risk-sharing by related companies (Khawaja and Mian 2005b)? To understand the importance of public information disclosure more investigation is needed.

Ultimately, complex interfirm relationships can alter a borrower's incentives and ability to repay a loan—necessitating that a lender analyze the firm's creditworthiness and consider the firm's interfirm relationships. Public information about these relationships can reduce information asymmetries, both between a borrower and a lender, as well as between lenders themselves. But public information can have drawbacks: it may reduce a lender's incentive to procure information if the information will be later revealed by a credit registry and imprecise or noisy public information may lead to inefficient decisions.

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Appendix A



STATE BANK OF PAKISTAN
BANKING SUPERVISION DEPARTMENT
CREDIT INFORMATION BUREAU
KARACHI

No.BSD/SU-62/201/ 20040908123203 Date 08/09/2004

REPORT REQUESTED BY:

AMOUNT BORROWED (indicated by blue arrow)

OVERDUE LOANS (indicated by blue arrow)

FIRM INFORMATION (indicated by blue arrow)

ACCESSED FROM

CREDIT WORTHINESS REPORT

CODE		BORROWER NAME		
PRESENT ADDRESS 				
PREVIOUS ADDRESS (IF ANY) N/A				
STATUS ☐ INDIVIDUAL ☒ COMPANY ☐ PARTNERSHIP ☐ NON-GROUP ☒ GROUP ☐ PUBLIC ☒ PRIVATE				
Consolidated Credit Exposure as on July 31, 2004 (Amounts are in '000' Rs.)				
Outstanding Liabilities		3,806,264	Recoveries During the Month 110	
Fund Based		3,065,417	Amount Under Litigation 0	
Non Fund Based		740,847	Write-offs (During last five years) N/A	
Overdues		0	No of times Rescheduling/Restructuring (During last five years) N/A	
Past Due 90 days		0		
Past Due 365 days		0		
Group Liabilities				
Outstanding Liabilities		Overdues Position (For Companies wise detail Pl. See Page 2)		
Total group borrowings		18,478,602	Past Due 90 Days 0	Past Due 365 Days 0
Details of Directors provided at the time of request				
Name	N.I.C Number	Name	N.I.C Number	
- NA				
Remarks				
List of directors for the firm				
Group Information (indicated by red arrow) Group loans overdue (indicated by red arrow)				

Fig. 9 Credit Worthiness Report from 2004. A copy of the information provided by the SBP about a prospective borrower in 2004.



STATE BANK OF PAKISTAN
CONSUMER PROTECTION DEPARTMENT
Corporate Credit Information Report

Report Ref. [REDACTED]		Report Date: 06/12/2012	
Corporate Profile			
Code: [REDACTED]		Name: [REDACTED]	
Present Address:		[REDACTED]	
Previous Address:		[REDACTED]	
Consolidated Credit Exposure As on 31/10/2012			
Outstanding Liabilities			
Fund Based	34,316,277	Amount Under Litigation	0
Non Fund Based	0	Writes-offs (During last ten years)	0
Overdues			
Past Dues 90 Days	0	Date and amount of Recovery	0
Past Dues 365 Days	0	No of times Rescheduling/ Restructuring During last five years	1
Group Entities of the Borrower			
Code #		Name Of Entity	
Group Information missing			
Credit Enquiries			
Enquiring Financial Institute			Enquiry Date
PRIVATE SECTOR COMMERCIAL BANK			22/11/12
PRIVATE SECTOR COMMERCIAL BANK			28/08/12
PRIVATE SECTOR COMMERCIAL BANK			31/05/12
PRIVATE SECTOR COMMERCIAL BANK			28/02/12
PRIVATE SECTOR COMMERCIAL BANK			02/12/11
Remarks			

Disclaimer:

The Information contained in this report has been compiled from the data provided by the financial institutions and does not represent the opinion of State Bank of Pakistan as to credit worthiness of the subject. Hence State Bank of Pakistan cannot assure any liability to the accuracy or completeness of the information. The information contained in this report is supplied on a confidential basis to you and shall not be disclosed to any other person

Fig. 10 Credit Worthiness Report from 2012. A copy of the information provided by the SBP about a prospective borrower in 2012.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jdeveco.2019.102407>.

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